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Hydrogen storage in metal hydride tanks equipped with metal foam heat exchanger

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ABSTRACT

This paper presents a two-dimensional mathematical model to evaluate transient heat and mass transfer in a metal hydride tank (hereinafter MHT) with metal foam heat exchanger. The model is validated by comparison with experimental data. A good agreement is obtained.

A study of the geometric and the operating parameters of the metal foam (such as base material, pore size and relative density) has been carried out to identify their influence on the performance of the charging process of the MHT. After that, the model was used as a tool to predict the performance of the hydrogen-charging process of two configurations of MHTs with metal foam heat exchangers. The obtained results showed that the reduction of the storage time due to the metal foam heat exchanger can be more important when the MHT is equipped with a concentric heat exchanger tube.

1. Introduction

The simultaneous growth of the world population and of air-polluting emissions produced by carbonaceous fuels imposes the replacement of gasoline by cleaner, renewable fuels such as hydrogen, which is especially attractive for fuel cell vehicle. The need for a worldwide conversion from fossil fuels to hydrogen requires the elimination of several barriers imposed along the different steps involved in hydrogen technology [1]. One of the main problems in large-scale usage of hydrogen energy in automotive industry is the storage process. Hydrogen forms metal hydrides with some metals and alloys leading to solid-state storage under moderate temperature and pressure that gives them the important safety advantage over the gas and liquid storage methods. Metal hydrides have higher hydrogen-storage density (6.5 H atoms/cm^3 for MgH_2) than hydrogen gas (0.99 H atoms/cm^3) or liquid hydrogen (4.2 H atoms/cm^3). Hence, metal hydride storage is a safe, volume-efficient storage method for onboard vehicle applications [2].

In fuel cell applications where the power source is mobile, such as automotive or portable usage, a hydrogen-carrying fuel reservoir is needed. The major bottleneck in the use of hydrogen-fuelled vehicles is the onboard hydrogen storage and the metal hydrides that are potential candidates for storing hydrogen [3].

However, it is known that the performance of hydrogen charge to/discharge from metal hydride depends on the heat transfer, mass transfer and kinetic reactions of the metal hydride bed. So, it is necessary for the design of optimized tanks to predict transient heat and mass transfer in the metal hydride bed. Heat exchangers are often employed in the literature to accelerate absorption/desorption rate in metal hydride tank. Hence, there have been numerous experimental

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Nomenclature

A	area, m^2
C_a	material constant, s^{-1}
C_p	specific heat, $J\,kg^{-1}\,K^{-1}$
E_a	activation energy, $J\,mol^{-1}$
h	heat transfer coefficient, $W\,m^{-2}\,K^{-1}$
H/M	hydrogen-to-metal atomic ratio
m	hydrogen mass absorbed, kg
M	molecular weight, $kg\,mol^{-1}$
N	number of moles of hydrogen gas
P	pressure, Pa
r	radial coordinate, m
R	universal gas constant, $J\,mol^{-1}\,K^{-1}$
t	time, s
T	temperature, K
Z	axial coordinate, m

Greek letters

ΔH	reaction heat of formation, $J\,kg^{-1}$
ε	porosity
λ	thermal conductivity, $W\,m^{-1}\,K^{-1}$
ρ	density, $kg\,m^{-3}$

Subscripts

a	absorption
e	equilibrium
eff	effective
f	fluid
g	gas
0	initial
ref	reference
s	solid
ss	saturated
t	total

and theoretical investigations on several aspects of hydride storage [4–14].

Spiral type heat exchanger for an MHT is developed by the authors [4]. The experimental results showed that when a heat exchanger is used, the charge/discharge times of the tank are considerably reduced. Moreover, the effects of different parameters such as cooling fluid temperature and mass flow, tank volume and operation pressure were investigated and it was concluded that a good choice of these parameters is necessary. An experimental setup is designed by authors [5] to study H_2 absorption and desorption processes and the effect of different heat exchangers on heat transfer from/to the hydride bed. In addition to that, a mathematical model is developed to predict the evolutions of temperature, pressure and hydrogen concentration inside an MHT. On the one hand, this model considers both the coupled heat and mass transfer inside the hydride bed. On the other, it considers the heating/cooling fluid. A computer program based on this model has been developed and used to study three configurations of MHTs. The calculated results are compared to experimental data and it is found that the proposed model adequately captures the main physical requirements of the sorption processes and can be employed for the design of a better MHT. Askri et al. [6] established a model to study the dynamic behaviour inside various designs of MHTs and optimization results indicate that almost 80% improvement of the storage time can be achieved when the design includes a concentric heat exchanger tube equipped with fins and filled with flowing cooling fluid. The optimization of hydrogen storage in metal hydride beds was investigated by Eustathios et al. [7], two novel cooling design options are investigated by introducing additional heat exchangers at a concentric inner tube and an annular ring inside the tank. Kaplan [8] studied the effects of heat transfer mechanisms on the charging process in metal hydride reactors under various charging pressures. Three different cylindrical reactors with the same base dimensions are designed and manufactured. The experimental results showed that the

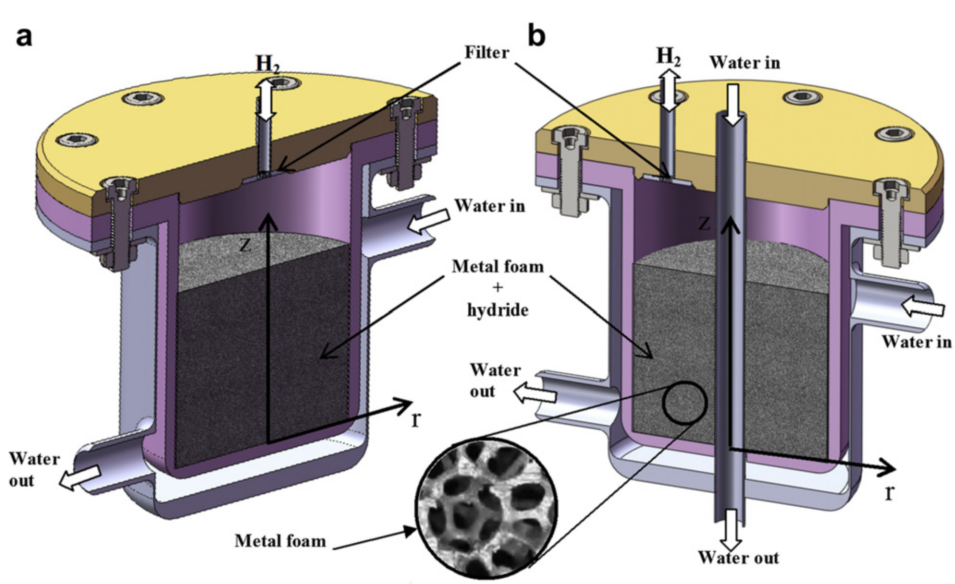

Fig. 1 – Schematic of MHTs used in simulations.

Table 1 – Parameters of MHTs used in simulations.

Parameter	Case 1	Case 2
Radius r (m)	0.04	0.04
Height H (m)	0.05	0.06
Concentric tube radius r_1 (m)	–	0.003

charging of MHTs is mainly heat transfer-dependent and the reactor with better cooling exhibits the fastest charging characteristics. Veerraju and Ram Gopal [9] presented a transient, two-dimensional model to predict the heat and mass transfer of elliptical metal hydride tubes and tube banks. Results obtained on elliptical hydride tube banks are compared with standard circular tube banks. It is shown that compared to a circular tube of same internal hydride volume, the use of elliptical hydride tube can provide more compact reactors with lower ratio of fan work to heat transfer. Hence, they may be suitable for applications such as storage of hydrogen for automobiles. On the other hand, Mc Donald and Rowe [10] focused on the external heat transfer improvements on metal hydride storage tanks by investigating the effects of addition of fins. A one-dimensional resistive analysis and two-dimensional transient model were used to determine the impact of external fins on the ability of a metal hydride tank to deliver hydrogen at a specified flow rate. It was found that the fins have a large impact on the pressure of the hydrogen gas within the tank when a periodic hydrogen demand is imposed, and their model results suggest that the metal hydride alloy at the center of the tank can be removed to reduce weight and cost, without detrimental effects to the performance of the systems. Jemni and Ben Nasrallah [11,12] examined two-dimensional heat and mass transfer within metal-hydrogen reactor for both absorption and desorption processes using two mathematical models. Laurencelle and Goyette [13] investigated transient heat and mass transfer within the reactor using a one-dimensional model to evaluate various designs of metal hydride reactors with planar, cylindrical or spherical geometry. They examined experimentally the hydrogen sorption in the reactor equipped with aluminium foam. The impact of the foam cell size has been studied. According to the model, the use of aluminium foam allows the reactor diameter to be increased by 7.5 times, without losing its performance.

All these studies have shown that the charge and discharge times of MHT strongly depend on the successful heat transfer from/to the hydride bed. Thus, different attempts to improve the effective thermal conductivity of metal hydride beds have been made: insertion of aluminium foam [14], integration of copper wire net structure [15], packing in a multilayer waved sheet structure, micro-encapsulated metal hydride compact [16], and compacting metal hydride powder with an expanded graphite [17]. The integration of heat exchangers represents an important solution to improve charge/discharge processes.

Heat exchangers have been widely studied in the literature, mainly to understand the heat and mass transfer mechanisms. Even though this can yield good heat and mass transfer performance, it is poor from the overall system weight point of view. For mobile applications requiring large hydrogen storage quantities, low overall system weight is an essential requirement. A well-designed heat exchanger configuration provides the possibility to optimize the system weight for given specifications. In the light of the above, a parametric study is necessary of an MHT with heat exchanger highlighting the influence of important geometric and operational parameters on the charging performance of the MHT. As a new material, metal foams are attracting increasing attention for a variety of applications. The advantages of metal foams lie in their low-density, large surface area in a limited volume and high-strength structure. Due to high manufacturing costs, metal foams have until recently been mainly used in the aerospace, ship-building and defence industries. More recently, the development of the metal-sintering method for foam manufacture has led to decreasing manufacturing costs and an increasing range of applications including heat and mass transfer [18]. The open-cell metal foam structure can be the desirable quality of a well-designed heat exchanger for MHT, i.e. a high specific solid-hydride interface surface area, good thermally conducting solid phase. Depending on the particular open-cell metal foam configuration, its specific surface area varies between approximately 500 and over 10,000 m^2/m^3 in compressed form [19]. The metal matrix can be manufactured from a high thermally conducting solid such as aluminium, copper and zinc, which, merely by its presence in an MHT, dramatically increases the overall effective thermal conductivity of the bed.

The present work seeks to develop a 2-D model of the hydriding process coupled with mass/heat transport in an MHT and to explore these issues that are extremely important for efficient hydrogen storage. The effect of metal foam exchangers on the charging process is evaluated. The impacts of the foam properties on heat transfer rate and hydrogen storage time are studied. Furthermore, novel cooling design options are investigated by introducing a concentric heat exchanger tube equipped with aluminium foam and filled with flowing cooling fluid.

2. Mathematical model

The different MHTs examined in this work are presented in Fig. 1. The hydride material inside the tanks is assumed to have the properties of $LaNi_5$–H_2. Two designs are considered:

Table 2 – The equilibrium pressure polynomial function coefficients.

Coefficients	a_0	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9
Absorption	0.0075	15.2935	−34.577	39.9926	−26.7998	11.03397	−2.841	0.446	−0.039	0.0014

Table 3 – Physical properties of the components of the MHTs.

	$LaNi_5$	Al foam	Hydrogen
Density ($kg\,m^{-3}$)	8300	2700	0.0838
Specific heat, C_p ($J\,kg^{-1}\,K^{-1}$)	419	963	14.89
Thermal conductivity, λ ($W\,m^{-1}\,K^{-1}$)	2.4	10.9	0.24
Porosity	0.55	0.91	–

- Case 1: a cylindrical tank equipped with metal foam that exchanges heat through its lateral and base surfaces at a constant temperature (Fig. 1a).
- Case 2: similar to case 1, with the addition of a concentric heat exchanger tube with metal foam and filled with flowing cooling fluid (Fig. 1b).

The dimensions of these MHTs are presented in Table 1.

Being axisymmetric, the MHTs of Fig. 1 are modelled as a two-dimensional axisymmetric system. For the hydride powder, the macroscopic differential equations are obtained by taking the average of microscopic equations over a representative volume and using the following assumptions:

- Hydrogen is assumed as an ideal gas as the pressure within the bed is moderate.
- The local thermal equilibrium is valid.
- The radiative heat transfer is negligible.
- The advection transport term is negligible.
- Thermo-physical properties are considered independent of bed temperature and concentration.

Considering these assumptions, the equations governing the heat and mass transfer and chemical reaction within the MHTs (Fig. 1) are given as follows [20].

2.1. Energy equation

Assuming thermal equilibrium between hydride and hydrogen, the energy equation can be expressed as:

$$(\rho C_p)_{eff}\frac{\partial T}{\partial t} = \frac{\lambda_{eff}}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \lambda_{eff}\frac{\partial^2 T}{\partial z^2} + \dot{m}\Delta H + \dot{m}T\left(C_{p_g} - C_{p_s}\right) \quad (1)$$

Fig. 2 – MHT used for model validation, taken for [13].

The coefficient $(\rho C_p)_{eff}$ in the heat transfer equations is a composite parameter:

$$(\rho C_p)_{eff} = \rho_{Foam} C_{p\,Foam} + (\rho_M + \rho_H) C_{p\,MH} + \rho_{H_2} C_{p\,H_2} \quad (2)$$

The density of each material present in the reactor is weighed taking into account the porosities ε.

$$\rho_{Foam} = (1 - \varepsilon_{Foam})\rho_{Foam[Bulk]} \quad (3)$$

$$\rho_M = \varepsilon_{Foam}(1 - \epsilon_{MH})\rho_{M[Bulk]} \quad (4)$$

$$\rho_H = x_H C_{MH} \rho_M \quad (5)$$

$$\rho_{H_2} = \varepsilon_{Foam}\varepsilon_{MH} M_{H_2}\frac{P}{RT} \quad (6)$$

where, x_H which can take values between 0 and 1, is the fraction of the maximal hydrogen loading capacity C_{MH} in a given mesh. M_{H_2} is the molecular mass of H_2.

The effective thermal conductivity is

$$\lambda_{eff} = \lambda_{Foam} + \epsilon_{Foam}\lambda_{MH} + \epsilon_{Foam}\epsilon_{MH}\lambda_{H_2} \quad (7)$$

2.2. Kinetic reaction

The hydrogen mass absorbed per unit time and unit volume, $\dot{m}$, is given by the following expression:

$$\dot{m} = C_a \exp\left(-\frac{E_a}{RT}\right)\ln\left(\frac{P}{P_e}\right)\frac{\rho_s}{M_s}\left\{\left(\frac{H}{M}\right)_{ss} - \frac{H}{M}\right\} \quad (8)$$

where P_e represents the equilibrium pressure that is given as a function of temperature and the hydrogen-to-metal atomic ratio (H/M).

$$P_e = f\left(\frac{H}{M}\right)\exp\left(\frac{\Delta H}{R_g}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right) \quad (9)$$

where $f(H/M)$ is the equilibrium pressure at the reference temperature T_{ref}; $f(H/M)$is a polynomial function of order 9,

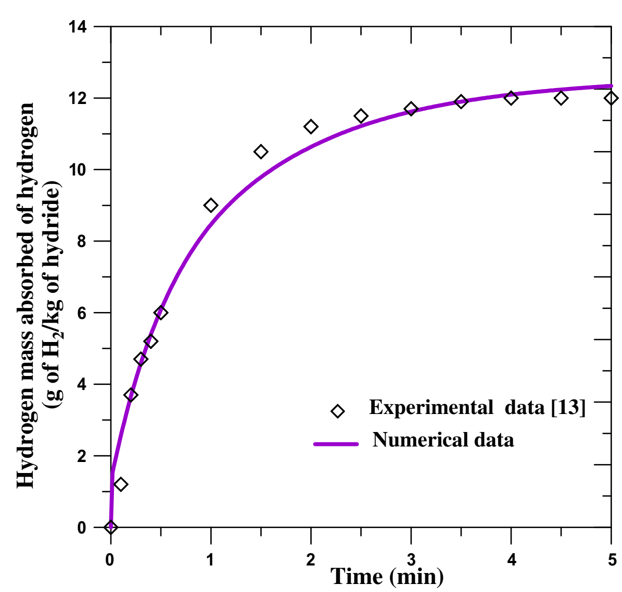

Fig. 3 – Calculated and measured hydrogen mass absorbed.

whose coefficients, for the absorption process, are given in Table 2 [20].

2.3. *Initial and boundary conditions*

- Initial conditions

Initially the hydride bed is assumed to be at a constant temperature and pressure:

$$P(r,z,0) = P_0, \quad T(r,z,0) = T_0 \tag{10}$$

- Boundary conditions

The boundary conditions depend on MHT configuration.

Case 1

The boundary conditions are expressed as:

$$\text{at } r = 0 \quad \frac{\partial T}{\partial r}(0,z,t) = 0 \tag{11}$$

$$\text{at } z = 0 \quad \lambda_{eff}\frac{\partial T}{\partial z}(0,r,t) = h_f \cdot (T - T_f) \tag{12}$$

$$\text{at } r = R \quad -\lambda_{eff}\frac{\partial T}{\partial r}(z,R,t) = h_f \cdot (T - T_f) \tag{13}$$

$$\text{at } z = H \quad -\lambda_{eff}\frac{\partial T}{\partial z}(H,r,t) = h \cdot (T - T_\infty) \tag{14}$$

where$h$ and h_f are the heat transfer coefficients between hydride bed/hydrogen gas and hydride bed/cooling fluid respectively.

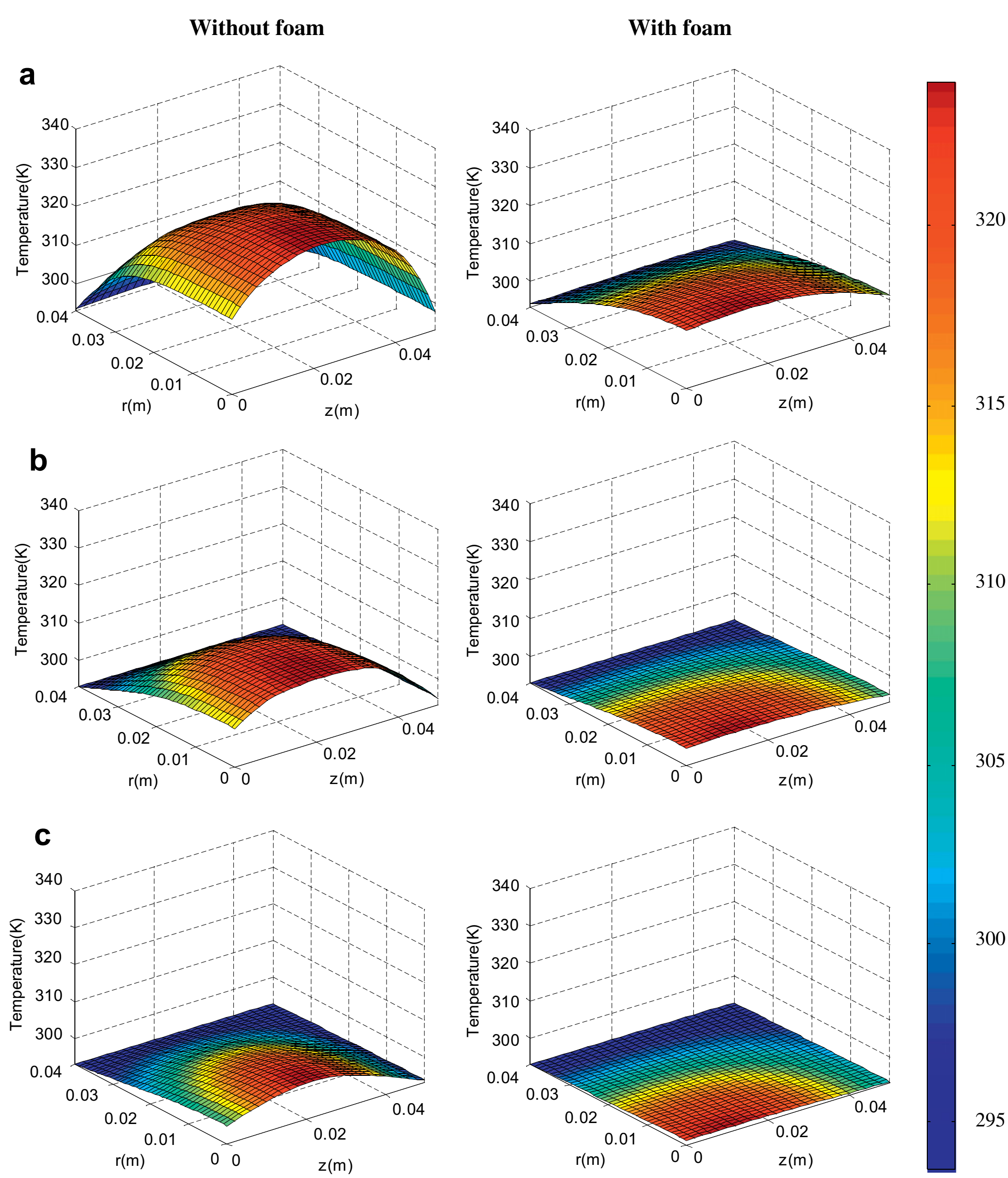

Fig. 4 – Time space evolution of hydride temperature: (a) 5 min, (b) 10 min, (c) 15 min.

Case 2

In this case, additional boundary conditions are imposed at r_1:

$$\text{at } r = r \quad -\lambda_{\text{eff}}\frac{\partial T}{\partial r}(z, r_1, t) = h\cdot(T - T_f) \tag{15}$$

where r_1 is the concentric tube radius.

The thermo-physical properties of materials used in computations are indicated in Table 3.

3. Model validation

In order to validate the established model, the numerical results are compared with the experimental results reported by Laurencelle and Goyette [13]. The MHT used for the experimental tests is described in Fig. 2. It is of a cylindrical configuration, with the innermost volume occupied by aluminium foam heat exchanger, and the intermediate annular region filled with 25 g of $LaNi_5$. Other information about the original experimental setup can be found in the work of Laurencelle and Goyette [13].

The predicted and the measured hydrogen storage capacity are plotted in Fig. 3. It is seen that numerical results agree satisfactorily with experimental data [13].

4. Results and discussion

To study the effect of metal foam on hydrogen storage time, MHTs with and without foam are considered. For the two MHTs, the same amount of hydride is used. In the present study, aluminium foam was inserted in the hydride bed with the specific purpose of enhancing the effective thermal conductivity of the system. To analyse the impacts of this aluminium foam exchanger on heat transfer, the temperature distribution in the MHT was determined and plotted in Fig. 4. From this figure, we noticed that at the same time, we obtained an important gradient temperature in the bed without foam compared to with foam. This result indicates that metal foam accelerates heat transfer from hydride power to cooling fluid.

Fig. 5 shows the radial evolution of the hydrogen concentration at $z = 0.025$ m. Near the wall, this concentration rate is larger than the interior of the bed. On the other hand, in the interior of the MHT, without foam, the concentration is smaller and the reaction proceeds slowly. This shows that the reaction can proceed just as the heat is removed by heat conduction in the interior of the bed.

Fig. 6 shows that 60% improvement of the time required for 90% storage can be achieved with the case without metal foam.

4.1. Effects of metal foam properties

Duocel Aluminium foam available in standard pore sizes includes 10, 20 and 40 pores per linear inch (PPI) and can be adjusted independently or by varying the relative density [21]. Foam properties can be tailored for a specific application and material response by adjusting the foam density, pore size, metals, and ligament structure (see Fig. 7).

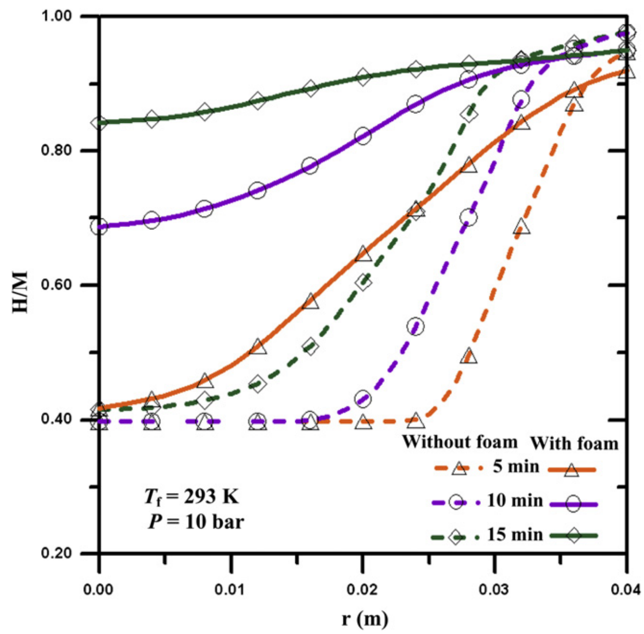

Fig. 5 – Evolution of hydrogen concentration in the hydride bed.

Duocel Aluminium foam from ERG Materials and Aerospace Corporation [21] is press-fitted inside the different cases of MHTs. This foam has a random open-cell structure (Fig. 7).

Performance of MHT with metal foam exchanger demands the definition of three major parameters, namely base material, pore size and the relative density.

4.1.1. Effects of base material

The properties of the base material are what determines all of the physical properties of the resulting foam such as specific heat and thermal conductivity.

To better understand the impacts of the integration of metal foam exchangers inside the hydride bed (case 1), the

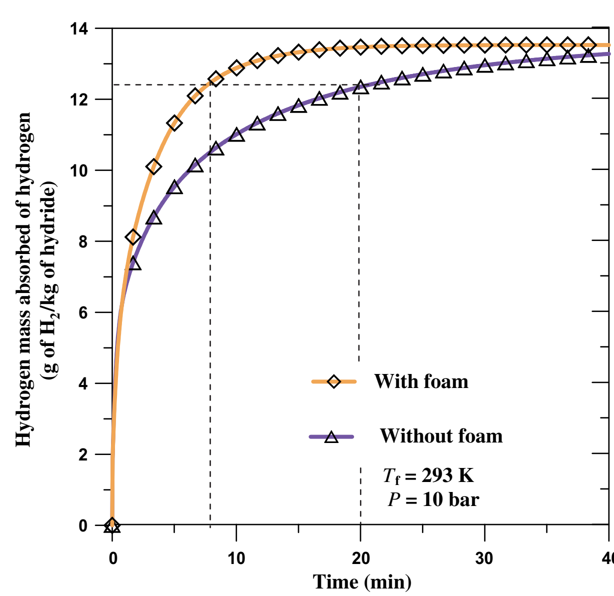

Fig. 6 – Effect of metal foam exchanger on the storage time.

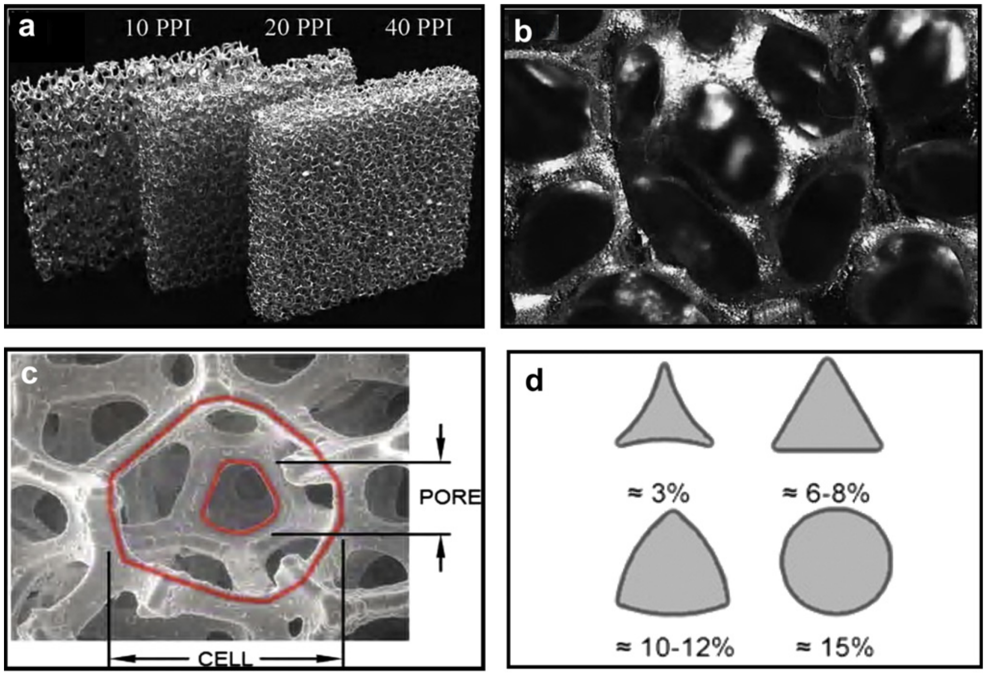

Fig. 7 – (a) Aluminium open-cell foams, (b, c) typical pore structure of open-cell metal foam and (d) ligament cross sections, taken from [21,22].

time evolution of the hydrogen mass absorbed for three materials of metal foam (zinc, copper and aluminium) are plotted in Fig. 8.

Fig. 8 shows that the metal matrix manufactured from aluminium solid has better hydriding performance compared to the other base material (copper or zinc). Therefore, the aluminium foam has the highest specific heat transfer coefficient.

4.1.2. *Effects of pore size*

Foam pore size defines how finely the raw material of a foam is divided. The bubble and strut structural shape is always constant, but 5 pores per inch (PPI) will visually appear more open and coarse than a 40 or 100 PPI foam. Accordingly, the foam pore size directly affects nominal ligament length and cross-section size, and pore diameter (see Fig. 7c). In turn, these micro-structural features influence specific surface area, thermal conductivity and heat transmission [21]. Pore size is an important geometric parameter that influences the performance of any metal foam exchanger. The pore diameter of foam was varied from 0.125 to 0.35 mm while keeping the other parameters constant, particularly the amount of metal hydride. Fig. 9 shows that the rate at which the hydrogen absorbed in the hydride bed with 0.125 mm pore diameter foam is rapid compared to the other pore diameters. Decreasing pore diameter leads to higher heat transfer surface area per unit available volume of the container, which causes higher heat transfer rates and thereby better hydriding performance.

4.1.3. *Effects of relative density*

While pore size controls the number and nominal size of the foam ligaments, the relative density controls the ligament cross-section shape and actual size (see Fig. 7c). Since foams can be compared to miniature three-dimensional truss structures, it is apparent that the cross section and moment of inertia of the struts or ligaments are a primary driver of foam mechanical properties like stiffness, crush strength, and thermal conductivity. Accordingly, relative density is the only foam characteristic that controls the thermal conductivity of heat exchangers [21–22].

To evaluate the foam density effect, three values are tested (3, 8 and 12%). Moreover, in this case, the same amount of metal hydride was used. Fig. 10 shows that the rate at which hydrogen is absorbed in 12% of density foam is faster

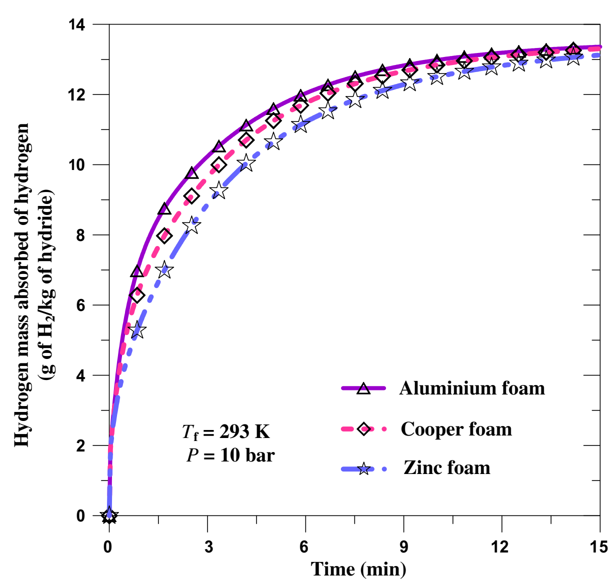

Fig. 8 – Effect of base material on the storage time.

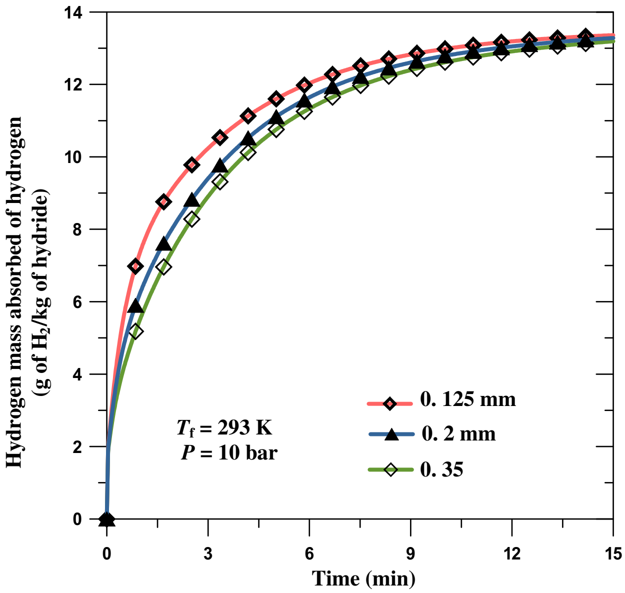

Fig. 9 – Effect of pore size on total H_2 mass stored.

compared to the other values. Increasing relative density leads to higher heat transfer surface area per unit available volume of the container. In addition to this, it is noticed that the hydride powder reaches the cooling fluid very quickly accompanied by faster heat transfer rate.

4.2. *Effects of MHT configuration*

To compare the considered cases (Fig. 1), the amount of hydride metal, the parameter values indicated in Table 1 and the boundary conditions were kept the same.

To evaluate the impacts of the integration of a concentric tube heat exchanger inside the hydride bed (Fig. 1b), the time evolution of the hydrogen mass absorbed for the two MHTs (Fig. 1) is presented in Fig. 11.

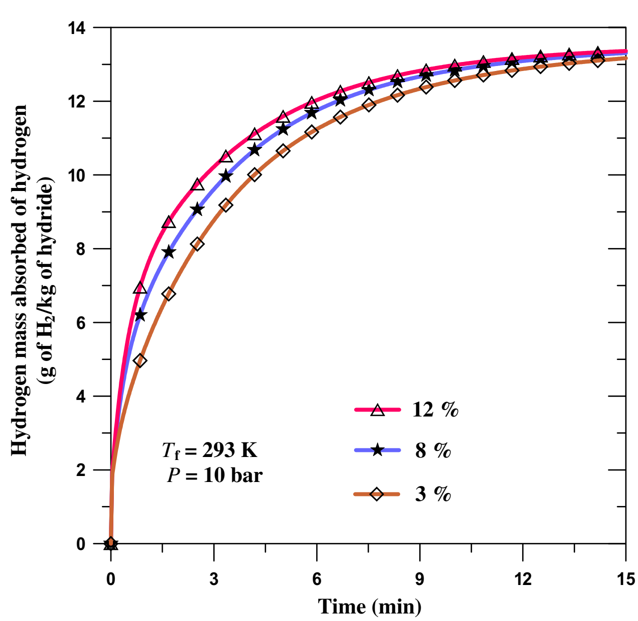

Fig. 10 – Effect of density foam on the storage time.

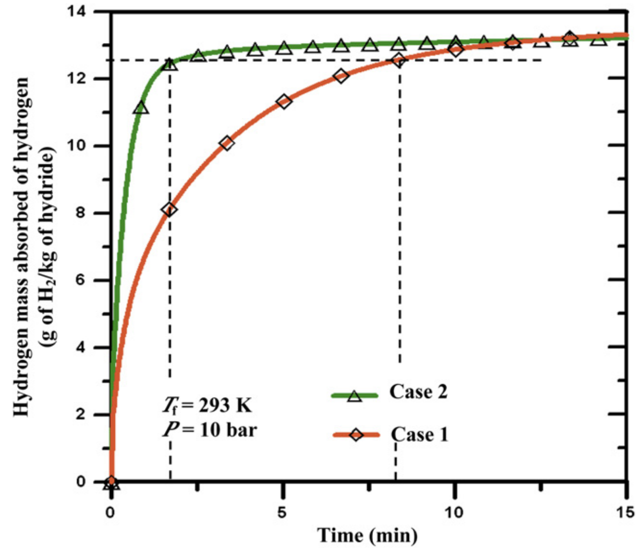

Fig. 11 – Time evolution of hydrogen mass absorbed.

As depicted in Fig. 11, the time required for 90% storage drops to about 2 min in case 2 representing a 75% improvement over the base case 1.

These results indicate that the integration of a concentric tube equipped with aluminium foam improves considerably the storage process.

5. Conclusion

A 2-D theoretical model has been developed describing hydrogen storage in metal hydride tanks (MHTs) with metal foam exchangers. Based on this model, a computer program has been implemented on Fortran 90 and applied to simulate the dynamic behaviour of a real system equipped with aluminium foam and containing the alloy of $LaNi_5$. Results for the hydrogen-charging process have been obtained and compared to experimental data; the good accord between numerical and experimental values shows that the proposed model is able to predict correctly the dynamic features of the storage system. The effects of various metal foam parameters such as base material, pore size and density of foam and configuration of heat exchanger on the hydrogen storage performance of the tank are studied. The obtained results reveal that the use of aluminium foam enhances heat transfer and consequently 60% improvement of the time required for 90% storage can be achieved compared to the case without metal foam. Also, it has been shown that, to effectively use metal foam exchangers in MHT, it is necessary to add a concentric heat exchanger tube to the MHT.

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